

Problem Set 4

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1. See file: `ProblemSet4_Fichtner_code.py`

2. See file:

(a) `test_q2.py`

(b) `test_q2.py`

3. (a) Regression model:

$$\ln(w_{i,t}) = \alpha + \beta_1 \text{Educ}_{i,t} + \beta_2 \text{Age}_{i,t} + \beta_3 \text{Age}_{i,t}^2 + \beta_4 \text{Black}_{i,t} + \beta_5 \text{Hispanic}_{i,t} + \beta_6 \text{OtherRace}_{i,t} + \varepsilon_{i,t},$$

$$\varepsilon_{i,t} \sim \mathcal{N}(0, \sigma^2).$$

(b) Likelihood Function:

$$L(\alpha, \beta_1, \dots, \beta_6, \sigma \mid y, X) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi} \sigma} \exp\left(-\frac{1}{2\sigma^2} (y_i - x_i^\top \beta)^2\right),$$

(c) Log-likelihood Function:

$$\ell(\beta, \sigma \mid y, X) = -\frac{n}{2} \log(2\pi) - n \log(\sigma) - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - x_i^\top \beta)^2.$$

(d) Negative Log-likelihood Function:

$$\text{NLL}(\beta, \sigma) = n \log(\sigma) + \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - x_i^\top \beta)^2.$$

For the python code, see file: `ProblemSet4_Fichtner_code.py`

4. Interpretation of β_1 (hysred: years of education) — MLE

Year 1971: $\beta_1 = 0.0665$

Year 1980: $\beta_1 = 0.0660$

Year 1990: $\beta_1 = 0.0955$

Year 2000: $\beta_1 = 0.1103$

Interpretation of β_1 (Education). In 1971, among male heads of household ages 25–60, one additional year of schooling is associated with an approximate **6.65% increase in wages**, on average, holding age and race constant. In 1980, 1990, and 2000, the estimated returns to education rise to **6.60%**, **9.55%**, and **11.03%**, respectively.

Over time, the wage premium to an additional year of schooling increased, suggesting that education became more strongly rewarded in the labor market.

5. See file: `test_q5.py`

6. Pytest Unit Test Passing Verification

```
(base) marlifirillo@Marlis-MBP CompEconPrivate % pytest -q
```

```
...  
3 passed in 24.82s
```

[100%]

```
(base) marlifirillo@Marlis-MBP CompEconPrivate % █
```